

$$\bar{C} = \left(\frac{2}{3}\right) C_{\text{root}} \frac{1 + \lambda + \lambda^2}{1 + \lambda} \quad (7.8)$$

$$\bar{Y} = \left(\frac{b}{6}\right) \left(\frac{1 + 2\lambda}{1 + \lambda}\right) \quad (7.9)$$

The mean aerodynamic chord (MAC or $\bar{C}$, pronounced “C-bar”) is the wing chord that acts like all of the area of the wing is compressed there. Recall that the wing subsonic pitching moments do not change with angle of attack if measured about a point 25% of chord back from the leading edge of the MAC. $\bar{Y}$ (“Y-bar”) is the spanwise location of C-bar.

For a vertical tail, the Y-bar value calculated in Eq. (7.9) must be doubled. This occurs because the total area is half the value obtained if the vertical tail were to be laid flat and converted to a symmetrical wing. All other calculations are the same as for a wing or horizontal tail.

Figure 7.24 also shows a quick method of determining C-bar and Y-bar from a trapezoidal wing. Y-bar is found as the intersection of the 50%-chord line and a line drawn from a point located at the tip chord length behind the root chord, to a point at the root chord length ahead of the tip chord. C-bar is found simply by drawing it at that location.

As mentioned in Chapter 4, the elliptically shaped wing has less drag than the trapezoidal wing already assumed, and if modern composite construction is used, it might no longer have the historically expected cost penalty. When designing an elliptical wing the span is found as usual from Eq. (7.5). The root chord C_{root} is found from Eq. (7.10a). Then the chord length is found as a function of distance Y from the centerline as in Eq. (7.10b). The MAC is 84.9% of C_{root} , and Y-bar is equal to 52.9% of the semispan.

$$C_{\text{root}} = \frac{4S}{\pi b} \quad (7.10a)$$

$$C(y) = C_{\text{root}} \sqrt{1 - \left(\frac{y}{b/2}\right)^2} \quad (7.10b)$$

Note that total area is $\pi/4$ times the product of span and root chord. Also, it is common for the chords of elliptical wings to be “slid” in the chordwise X direction so that the 25% of chord line is straight and unswept. This has no effect on the above calculations, but does move the 25% of MAC location a bit forward.

7.8.2 Wing Location with Respect to the Fuselage

The location and length of the MAC are important because the designer locates the wing on the aircraft so that some selected percent of the MAC is aligned with the aircraft center of gravity. This provides a first estimate of the wing position to attain the required stability characteristics.

If the airplane is a pure flying wing, with no other components than a wing of trapezoidal shape, it will be neutrally stable if its center of gravity is at that 25% of C -bar location. The pitching moment doesn't change around that point so that if the angle of attack changes, the moments do not change. This is the very definition of neutral stability.

An aft tail adds to the stability. For a stable aircraft with an aft tail, the wing should be initially located such that the aircraft center of gravity is at about 30% of the mean aerodynamic chord. When the combined effects of the fuselage and tail are considered, this gives a reasonable level of stability.

When designing an unstable aircraft with an aft tail such as the F-22, the wing must be farther forward. A good first approximation is to locate the wing such that the center of gravity is at about 40% of the mean aerodynamic chord.

For a canard aircraft, such rules of thumb are less reliable due to the canard downwash and its influence upon the wing. For a control-type canard with a computerized flight control system (i.e., an unstable aircraft), the wing can initially be placed such that the aircraft center of gravity is at about 15–20% of the wing's mean aerodynamic chord.

For a lifting-type canard, the mean aerodynamic chords of the wing and canard should both be determined, and a point at about the 15% MAC for each should be identified (20–25% for an unstable aircraft). Then the combined MAC location can be determined as the average of these percentage MAC locations for the wing and canard, weighted by their respective areas. Note that this is a very crude estimate!

After the initial layout is completed and analyzed using the methods of Chapters 12–19, the wing will probably need to be moved and the tails resized to meet all required stability and control characteristics. Hopefully the initial estimates will be close enough so that major changes will not be needed.

7.8.3 Wing/Tail Lofting

After positioning and drawing the trapezoidal wing and tails, the actual exposed wing and tails are designed. As changes are made, those original plan-forms should be retained on the drawing or as components in the CAD file, to facilitate their use in aerodynamic calculations. Recall that the trapezoidal wing is used as the reference wing for aerodynamic coefficients—don't forget what it is!

The trapezoidal wing is defined to the aircraft centerline and is based upon the projected area (i.e., dihedral does not affect the top view of the reference wing). The actual, exposed wing begins at the side of the fuselage and includes the effect of the dihedral upon the true-view area. The dihedral angle increases the actual wing area equivalent to dividing by the cosine of the dihedral angle.